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HEALTHTECH INNOVATION CHALLENGE 2025

Jointly hosted by HMFWD Dept, AP, APIS & RTIH



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**Innovate, Diagnose, Treat: Driving AI-Powered, Portable
Healthcare in Andhra Pradesh- AP Healthtech Innovation
Challenge 2025**

India's Disease Burden and the Healthcare Challenges Facing Citizens:

India today stands at a critical juncture in its public health journey. While the country has made impressive strides in reducing mortality, expanding primary healthcare, and strengthening national health programmes, the overall disease burden remains high and complex. Citizens face a **dual challenge**: the continuing threat of infectious diseases and the rapidly rising burden of non-communicable diseases. This “double burden” is further compounded by malnutrition, maternal and child health gaps, climate-linked health risks, and systemic limitations in last-mile health service delivery.

Andhra Pradesh faces a dual healthcare burden where rising non-communicable diseases—such as hypertension, diabetes, cardiac disease, cancer, and chronic kidney disease—coexist with persistent infectious diseases, maternal and child health gaps, and high levels of anaemia and malnutrition. Many of these conditions remain undetected until late stages due to limited access to diagnostics at the primary and community levels. Rural mandals, tribal regions, and coastal habitations often depend on distant district hospitals for even basic tests like creatinine, HbA1c, LFTs, and lipid profiles, leading to delays in diagnosis, high out-of-pocket expenditure, and missed opportunities for timely treatment.

The state's unique geography—spanning a long coastline vulnerable to cyclones, widespread rural settlements, and remote tribal belts—further complicates last-mile service delivery. Seasonal outbreaks of dengue, malaria, respiratory infections, and water-borne diseases add to the healthcare load. Specialist availability remains concentrated in urban centers, making follow-up care, chronic disease management, maternal risk monitoring, and emergency care difficult for citizens in underserved areas.

Given this backdrop, Andhra Pradesh urgently requires portable, affordable, and AI-enabled medical technologies that can strengthen PHCs, CHCs, and Health & Wellness Centres; support frontline workers; and bring diagnostics, monitoring, and tele-specialty care closer to citizens.

Department of Health, Medical and Family Welfare, Government of Andhra Pradesh in partnership with **AP Innovation Society & Ratan Tata Innovation Hub (RTIH)** launched the Healthtech Innovation Challenge 2025 aimed at identifying, evaluating, and integrating innovative Medical/Health technologies that enhance the efficiency and reach of healthcare services, particularly in field-based and resource-limited settings. It aligns with the government's broader vision of achieving **Universal Health Coverage (UHC)**.

1. Challenge Vision and Rationale

Overarching Goal: To enhance healthcare service delivery through the integration of cutting-edge, affordable, and portable medical technologies.

Rationale: Conventional healthcare models, reliant on centralized hospital-based care, are often unable to address the growing demand for quality medical services in geographically dispersed communities. Healthcare delivery in remote and underserved areas faces significant challenges, including inadequate infrastructure, a shortage of skilled medical professionals, and limited financial resources, which lead to diagnostic and treatment delays. This challenge seeks to leverage advancements in portable devices, digital health solutions, and telemedicine to redefine accessibility and enable early disease detection and remote patient monitoring.

Core Objectives:

1. **Identify Viable Medical Technologies** that address key healthcare challenges such as diagnostic delays and lack of timely treatment in underserved areas.

- 2. **Ensure Scalability and Integration** of proven technologies into the existing healthcare system for long-term sustainability.
- 3. **Foster Stakeholder Engagement and Collaboration** among technology providers, healthcare institutions, investors, and policymakers.
- 4. **Enhance Public-Private Partnerships** to drive investment and strengthen overall health infrastructure.

I. Targeted Participants and Eligibility

The challenge invites the entire spectrum of the healthcare innovation ecosystem to participate, including organizations with minimum three years of relevant industry experience.

Innovation Category	Description
Early to Growth-Stage Startups	Deep-tech startups focused on developing scalable, technology-driven healthcare solutions for the Indian market.
Independent Innovators & Researchers	Individual innovators or researchers who have developed a mature, ready-to-demonstrate device or software solution that can function effectively in resource-constrained settings.
Research & Academic institutions	University labs, research institutions, and technology incubators seeking a public platform to validate and pilot their cutting-edge inventions.
MedTech Manufacturers	Companies with existing, proven portable technologies ready for large-scale adoption and integration into public health systems.

II. Comprehensive Scope of MedTech Solutions (AI-First Focus)

The scope encompasses cost-effective and scalable medical technologies designed for seamless integration into existing healthcare settings.

Category	Challenge Area & Preferred Innovation Type
AI-Powered Diagnostic Tools	Solutions using machine learning and deep learning algorithms to analyse medical images, lab results, and patient histories for early disease detection , risk stratification, and real-time data-driven decision-making.
Portable Point-of-Care Testing	Compact, high-accuracy devices for critical tests that offer fast turnaround time and low infrastructure dependency. Priority tests include Serum Creatinine, Lipid Profile, LFTs, KFTs (assessing renal and hepatic function), and Complete Blood Picture (CBP).
Smart Monitoring & Wearables	Technologies enabling real-time, continuous tracking of vital parameters (e.g., ECG, blood pressure, oxygen saturation, glucose), with automated alert systems and synchronization with electronic health records.
Remote Care & Telemedicine	Virtual consultation platforms, remote diagnostic tools, and cloud-based patient data management designed to bridge geographical disparities and ensure timely medical interventions.
Non-Invasive Technologies	Technologies that enhance patient comfort and diagnostic efficiency, providing quick and reliable results for point-of-care applications.
High-Priority Public Health Focus	Solutions specifically impacting Maternal Health, Child Health, Malnutrition, Anaemia, TB, Blood Sugar, Renal function, and Cancer diagnostics.

For more details on the list of interventions may please refer to Annexure I

III. Key Evaluation and Selection Criteria

Proposals will be evaluated by a committee of healthcare and technology experts based on technical merit and strategic impact, focusing on the following criteria:

#	Valuation Parameter	Marks	Rationale
1.	Innovation, Uniqueness & Technical Differentiation	20	More weight because innovation drives transformative impact. Evaluates AI/tech novelty, differentiated approach, IP strength, etc.
2.	Scalability & Replicability Across Andhra Pradesh	20	Critical for state-wide deployment. Evaluates supply chain readiness, manufacturability, cloud/infra capability, and expansion feasibility.
3.	Applicability to Public Health Priorities (AP Context)	15	Measures how well the solution addresses AP's top burdens: NCD, MCH, CKD, TB, malnutrition, remote areas, etc.
4.	Usability & Field Readiness (Portability, Training, Simplicity)	15	Ensures frontline workers (ANMs/ASHAs/CHOs) can practically use the device/software.
5.	Regulatory Readiness & Safety Compliance	12	Ensures safety, validation, ICMR/CDSCO certifications, ethical approvals.
6.	Pilot Deployment Readiness (TRL Level, Demo Capability)	10	Measures whether the innovation can be quickly demonstrated and piloted.
7.	Cost-Effectiveness & Value for Money	8	Important but not dominant. Evaluates cost per test, affordability, maintenance expense.
TOTAL		100	

IV. Challenge Logistics and Scaling Pathway

Submission Requirements:

- **Format:** All proposals must be sent **electronically only and must be submitted through the online** <https://rtih.co.in/healthtech-innovation-challenge-2025>

Important Timelines:

Activity / Stage	Date
Release of Innovation Challenge Notification	13.11.2025
Opening of Application	14.11.2025
Last Date for Submissions	05.12.2025
Preliminary Screening	05.12.2025
Shortlist Announcement (Top 25–30 Technologies)	12.12.2025
Final Presentation /Demonstration	15.12.2025

For any questions please reach us at: +91 7042488871/ +91 905210852

Post-Demonstration Opportunity: Following successful demonstrations before senior government officials and key stakeholders, selected applicants will be invited for further deliberations focusing on practical aspects of integration, training requirements, and regulatory compliance. Based on these discussions, a structured roadmap will be developed to facilitate **pilot testing** and **potential large-scale adoption** of the most promising innovations across the healthcare system

Annexure I

List of major public-health issues, their key challenges, along with the types of innovative solutions required

#	Intervention	Key Challenges / Indicators	Solution Categories (MedTech, AI, Digital, System Innovations)
1	Maternal & Child Health (MCH)	- Maternal mortality - Neonatal mortality, IMR, U5MR - High-risk pregnancies undetected - Birth asphyxia, low birth-weight	• Portable foetal monitoring (dopplers, CTG) • AI-based antenatal risk scoring • Digital high-risk pregnancy tracking • Remote/tele-obstetrics • Portable neonatal monitoring (SpO₂, bilirubin) • Community-level screening/POCT kits
2	Malnutrition & Micronutrient Deficiencies	- High anaemia prevalence in women, adolescents, children - Stunting, wasting, underweight - Poor dietary diversity - Weak follow-up under ICDS	• Non-invasive Hb measurement devices • Portable biochemical POCTs (ferritin, vit D) • AI-based nutrition counselling apps • Smart anthropometry tools • Fortified food supply-chain tracking
3	Tuberculosis (TB) & Communicable Diseases	- Underdiagnosis & delayed detection - Drug-resistant TB - Poor treatment adherence - High burden of pneumonia, diarrheal diseases - Malaria & dengue outbreaks	• Portable NAAT / TB rapid molecular tests • AI-supported chest X-ray interpretation • Digital adherence (video DOT, smart pillbox) • Field-deployable dengue/malaria tests • Tele-infectious disease consults
4	Non-Communicable Diseases (NCDs)	- Cardiovascular diseases, diabetes, cancer - Hypertension, obesity - Low screening rates - Late diagnosis and costly complications	• Portable NCD POCT (HbA1c, lipids, creatinine) • BP/ECG/glucose wearables with alerts • AI-based risk predictors (CVD, diabetes) • Tele-cardiology & remote monitoring • Early cancer screening tools (oral, cervical,

			breast) • Lifestyle modification digital platforms
5	Renal & Hepatic Health (CKD, Liver Disease)	- Rising CKD burden - Low awareness and late diagnosis - Limited nephrologists - High diabetes/hypertension leading to CKD	• Portable creatinine & urine albumin POCT • AI kidney-risk scoring • Tele-nephrology hubs • Remote diet and medication monitoring apps • Portable liver function POCT (LFTs)
6	Respiratory Diseases & Air Pollution	- COPD, asthma, chronic respiratory illness - Pollution-related morbidity - High particulate exposure in cities & rural biomass use	• Portable spirometry devices • Air-quality IoT sensors • AI-based exposure-risk alerts • Tele-pulmonology services • Portable oxygen/respiratory triage kits
7	Mental Health & Neurological Disorders	- High prevalence of depression, anxiety, substance use - Stroke burden - Limited psychiatrists and neurologists - Stigma & low help-seeking	• Digital mental health screening apps • Tele-psychiatry platforms • AI triage for neurological symptoms • Remote rehabilitation applications • Crisis hotlines integrated with digital triage
8	Nutrition & Child Development	- Early childhood developmental delays - Poor growth monitoring - Low coverage of early interventions	• Digital growth charts • Smart MUAC/bioimpedance devices • Apps for caregiver education • Tele-consultation with paediatric specialists
9	Immunization & Preventable Diseases	- Dropouts in routine immunization - Cold chain disruptions - Outbreak-prone regions	• IoT-based vaccine cold-chain monitoring • Digital immunization records (ABDM-linked) • SMS/app-based reminders • Rapid antigen/POCT for outbreak detection
10	Vector-borne Diseases	- Malaria, dengue, chikungunya - Climate-linked outbreaks - Slow surveillance	• Portable malaria/dengue rapid tests • Predictive analytics for hotspot mapping

			• Drones for vector surveillance • Community rapid-response tools
11	Environmental & Climate-Linked Health Risks	- Heat stress - Water contamination - Climate-sensitive disease spikes	• IoT water-quality sensors • Heat-risk alert systems • Climate-disease dashboards integrated with CHCs
12	Injuries & Road Traffic Trauma	- High road accident burden - Delays in trauma care - Fragmented emergency systems	• Prehospital triage apps • GPS-linked ambulance dispatch • Portable trauma imaging (handheld USG) • Tele-emergency/tele-trauma support
13	Oral Health	- High prevalence of dental caries, oral cancer - Limited rural dental access	• Portable oral exam devices • AI-based oral cancer screening • Teledentistry platforms
14	Eye Health	- Cataract, diabetic retinopathy - Preventable vision impairment	• Portable fundus imaging • AI-based DR/cataract grading • Tele-ophthalmology screening camps
15	Geriatric & Palliative Care	- Increasing elderly population - Chronic disease + multimorbidity - Poor access to home-based care	• Remote monitoring kits for elderly • Home-based care digital platforms • Medication adherence tools
16	Antimicrobial Resistance (AMR)	- Overuse of antibiotics - Limited surveillance - Poor stewardship	• AMR surveillance platforms • Rapid diagnostics to reduce empirical prescribing • Decision-support for rational antibiotic use
17	Digital Health, Data & Interoperability (Cross-cutting)	- Fragmented systems - Incomplete patient records - Low data visibility	• ABDM-compliant health data exchange • Integrated dashboards • AI-assisted population health analytics
18	Workforce & Health System Gaps	- Shortage of specialists - Limited training for frontline workers - High workloads	• Tele-mentoring (ECHO model) platforms • Decision-support apps for ASHAs/ANMs • AR/VR-based medical training

19	Supply Chain & Health Logistics	- Stock-outs - Temperature-sensitive drugs - Inaccurate demand forecasting	• IoT-enabled stock tracking • Supply-chain dashboards • Drone-based medical logistics in hard-to-reach areas
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